

MUNGUIA-FLORES, HANNAH

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POST COVID-19 EXPANDED PANEL FINAL

Post COVID-19 Test Panels

Clinical findings are an important key in evaluating care and potential treatment pathways for those with Long COVID or Post COVID-19 conditions, as stated by the Centers for Disease Control (CDC). This panel of laboratory tests provides information for those with ongoing symptoms.

These tests are to help you and your doctor determine if your current symptoms are potentially due to a post-COVID-19 infection or if you have another underlying health problem causing your symptoms. They include:

C-Reactive Protein (Basic and Expanded panels):

The C-Reactive Protein (CRP) Test measures the amount of CRP in your blood. CRP is a protein produced by the liver and is a marker of inflammation. Increased CRP levels are found in inflammatory conditions including bacterial infection, rheumatic fever, potential malignancies, autoimmune disorders, active arthritis, cardiac disease, heart attack, and conditions related to the immune system. Those with severe lingering COVID-19 symptoms may have elevated CRP levels which could indicate higher risk of cardiovascular disease.

Thyroid function Test (Basic and Expanded panels):

The Thyroid Function Test measures thyroid-stimulating hormone (TSH) levels in the blood to evaluate thyroid function, which is critical for regulating weight, energy, body temperature, mood, and overall health. If your TSH levels are abnormally high or low, this may indicate that your thyroid is either underactive (hypothyroidism) or overactive (hyperthyroidism). If test results identify abnormal TSH levels, free thyroxine (T4) blood levels will be evaluated using your submitted sample to provide information needed to accurately diagnose hypothyroidism. There is a potential parallel between Long COVID and abnormal thyroid function which may trigger pre-existing thyroid disease or autoimmunity caused by a thyroid infection, however many thyroid conditions can be managed with treatment.

Vitamin B12 (Basic and Expanded panels):

The Vitamin B12 Test measures the level of vitamin B12 in your blood. Vitamin B12 is absorbed in the small intestines and then makes its way to the blood and cells. The body also stores vitamin B12 in the liver for future use. Vitamin B12 is essential in supporting the nervous system, making red blood cells, and creating DNA (the genetic material present in our cells). Vitamin B12 must be supplied by the diet or supplementation because the body is not able to produce it independently. It is naturally found in animal proteins, such as red meat, fish, poultry, eggs, and dairy. Plant foods are not a reliable source of vitamin B12 unless they are fortified. Low levels of vitamin B12 may occur if you do not consume enough vitamin B12 or the body does not store or absorb enough of the vitamin. People who are deficient in vitamin B12 may or may not have any symptoms of anemia (a condition in which your body does not have enough healthy red cells to carry oxygen to your body's tissues) or nerve damage. Early detection is important to prevent irreversible damage over time, especially to the nervous system. Vitamin B12 supports the immune system and helps your body fight off viruses. Many symptoms of long COVID are the same as those with a vitamin B12 deficiency. If your body is low in B12, you may not be able to fight viruses as well.

Vitamin D (Basic and Expanded panels):

The Vitamin D Test measures the amount of 25-hydroxyvitamin D, an indicator for the level of Vitamin D in your body. Vitamin D is involved in regulating calcium and phosphorus, which are essential nutrients to promote bone strength and cellular health. Vitamin D can also help the immune system fight infections. Your body produces Vitamin D after exposure to sunlight or from consuming certain foods and supplements. An adequate level of this vitamin is necessary for immune function, heart and blood vessel health, insulin regulation, and mood. Abnormal levels of vitamin D can indicate bone disease, nutrition disorders, or other metabolic complications. Many symptoms of Long Covid are the same as those with a Vitamin D deficiency. If your body is low in vitamin D, you may not be able to fight off viruses as well.

Urinalysis (Basic and Expanded panels):

A urinalysis, also called a urine panel, consists of a series of tests used to evaluate visual, chemical, and microscopic features of urine. Urinalysis may be used to identify potential risks for diabetes, liver disease, and kidney disease. A urinalysis can help to identify lower urinary tract symptoms and kidney injury caused by inflammation from a severe COVID-19 infection.

Complete Blood Count (CBC) (Expanded panel only):

A CBC measures 15 components of your blood to help identify bleeding disorders or infections and to evaluate abnormal levels of specific blood components, such as low levels of red blood cells, which transport oxygen to body cells and tissues (anemia) and high levels of abnormal white blood cells that crowd out red blood cells and platelets (leukemia). T cells destroy the body's own cells that have been infected with viruses or have developed cancer. Both B cells and T cells protect against future invasions of bacteria, viruses, and parasites by generating antibodies that retain antigen recognition and protect against such antigens. Red blood cells and T cells could be impacted due to a previous COVID-19 infection. The CBC can detect blood abnormalities.

Ferritin (Expanded panel only):

Ferritin is a protein in the blood that contains iron and helps you understand how much iron your body is storing. Blood levels of ferritin correlate to the body's iron status. Because iron is an important component of oxygen-carrying hemoglobin in the blood, low ferritin levels may indicate an iron deficiency. If your ferritin levels are higher than normal, you may have an underlying condition that causes your body to store too much iron. It could also indicate liver disease, rheumatoid arthritis, or inflammatory diseases, for example.

Ferritin, another protein, is the best indicator of the amount of uncommitted iron reserve that the body has in storage and is useful in the diagnosis of hypochromic microcytic anemia (low red blood cell count associated with small red blood cells).

Ferritin Blood Test

The ferritin test is ordered to assess a person's iron stores in the body.

The thyroid gland produces hormones that regulate or affect many bodily functions. Thyroid stimulating hormone (TSH) is the hormone which controls the thyroid gland function. An excess or deficiency of this hormone can affect energy levels, mood, and many functions. When the thyroid gland fails, due to primary disease of the thyroid, TSH levels increase. This condition is called primary hypothyroidism, meaning low functioning of the thyroid. In contrast, when the thyroid gland is overactive and producing too much thyroid hormone, the TSH level decreases. This is called primary hyperthyroidism, meaning excessive functioning of the thyroid. Both hypothyroidism and hyperthyroidism can be detected by the TSH test.

For pregnant women the reference ranges are:

First trimester 0.26-2.66

Second trimester 0.55-2.73

Third trimester 0.43-2.91

Question 1. What is TSH and how is it measured?

Thyroid stimulating hormone (TSH) is one of the most important hormones currently used to diagnose thyroid abnormalities. This glycoprotein is secreted by the pituitary and stimulates release of thyroxine (T4) and triiodothyronine (T3) from the thyroid gland. TSH release from the pituitary is controlled by thyrotropin releasing hormone (TRH) stimulation and negative feedback from free T3 and free T4.

Question 2. Does the time of day matter when sampling for TSH testing?

Yes. TSH concentration follows a diurnal rhythm. Typically, the peak occurs around midnight and the nadir (~50% of the peak value) around mid-day. Population-based reference intervals are generally obtained from subjects tested in the daytime, closer to the trough than to the peak. So, when evaluating a patient's serial TSH concentrations, differences in sample collection time should be considered.

Question 3. How variable is TSH?

TSH has moderate intraindividual variability and even more marked interindividual variability. The interindividual coefficient of variation is about 32%; consequently there is a wide population-based reference interval for TSH. Since the intraindividual variation is considerably less, comparing a specific patient's current TSH level with any past level may be more illuminating than comparing the patient's current TSH level to the reference interval. A difference of 0.7 mIU/L or greater is considered significant when evaluating a patient's serial TSH values.

Thyroid Screen - TSH

TSH refers to thyroid stimulating hormone. This hormone is produced in the pituitary gland and it acts on the thyroid gland in the front of your neck. Here it stimulates the production of thyroid hormones and their release into the blood. While high or low levels of TSH in the blood may indicate a thyroid disorder, additional tests may be ordered to better understand the specific medical condition. More information on TSH and screening for thyroid disorders may be found on WebMD by clicking here: <https://www.webmd.com/women/what-is-tsh-test#1>.

Thyroid Function Tests

The blood tests that are most widely used to evaluate thyroid function include those that measure TSH, T4, T3, free T4, and thyroid antibody levels. Read more about these tests in the brochure provided by the American Thyroid Association (ATA).

Download the brochure from the ATA website : http://www.thyroid.org/wp-content/uploads/patients/brochures/FunctionTests_brochure.pdf

C-Reactive Protein

C-reactive protein is an acute phase protein that is used to detect inflammation due to acute conditions or to monitor chronic conditions.

Vitamin B12, one of the B vitamins, is important for the synthesis or production of DNA, our core genetic material in each cell. Deficiency of vitamin B12 is most easily seen in quickly dividing cells such as our red blood cells. Deficiencies in vitamin B12 are reflected in increased size of the red blood cells as well as decreased production leading to anemia. Folate deficiency has the same effect so it is common to measure both vitamin B12 and folate to determine if one or both may be deficient and causing the changes in the red blood cells. Vitamin B12 is important for maintaining the effective function of the nervous system. Vitamin B12 is found in animal products such as meat, shellfish, milk, cheese and eggs. Strict vegetarians who do not eat animal products and young children of mothers who are strict vegetarians are at increased risk for developing vitamin B12 deficiency and the associated anemia. Supplemental vitamin B12 may be advised for such individuals.

Vitamin B12 Fact Sheet

In addition to keeping the body's nerve and blood cells healthy, the nutrient vitamin B12 helps make DNA, the genetic material in all cells.

Read more in this fact sheet provided by the NIH Office of Dietary Supplements : <http://ods.od.nih.gov/pdf/factsheets/VitaminB12-Consumer.pdf>

Your Results

Ferritin

Desired Range: 16-154 ng/mL



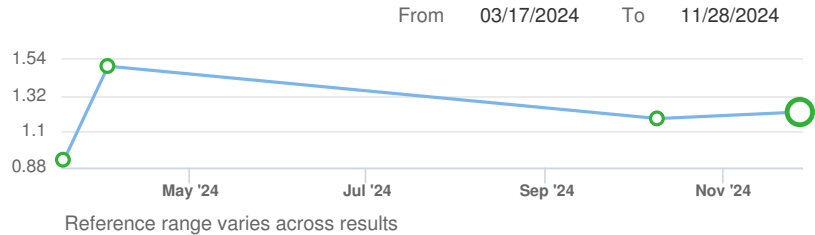
What is Ferritin?

Ferritin, a protein, is the best indicator of the amount of uncommitted iron reserve that the body has in storage and is useful in the diagnosis of hypochromic microcytic anemia (low red blood cell count associated with small red blood cells).

TSH W/REFLEX TO FT4

Unit of Measure: mIU/L

Result: 1.22



Reference Range

> or = 20 Years 0.40-4.50

Pregnancy Ranges

First trimester 0.26-2.66

Second trimester 0.55-2.73

Third trimester 0.43-2.91

C-REACTIVE PROTEIN

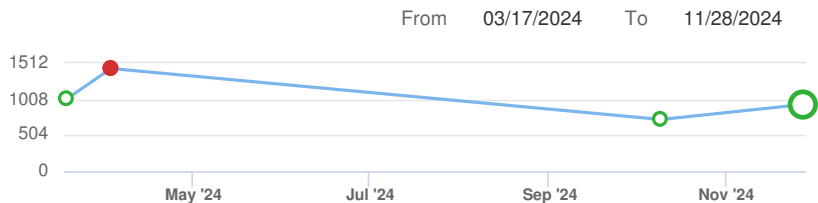
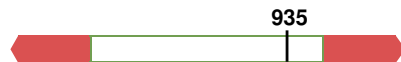
Desired Range: <8.0 mg/L

Result: <3.0



VITAMIN B12

Desired Range: 200-1100 pg/mL



What is Vitamin B12 (Cobalamin)?

The Vitamin B12 Test measures the level of vitamin B12, also known as cobalamin, in the blood. Cobalamin is an essential water-soluble vitamin that helps your body make red blood cells, create DNA (the genetic material present in our cells), and keeps your nervous system healthy.

If you don't eat enough cobalamin, or if your body doesn't store or absorb enough of it, you may have low levels. People who are deficient in cobalamin may not have any symptoms or may have symptoms of anemia, depression, weakness, tiredness, memory loss, or nerve damage. Early detection is important to prevent irreversible damage over time, especially to the nervous system.

Our bodies don't make cobalamin, so we need to get it daily from food or supplements. Cobalamin is naturally found in animal proteins and other foods such as red meat, fish, poultry, eggs, and dairy. Plant foods are not a reliable source of cobalamin unless they are fortified. Vegetarian sources of Vitamin B12 are found in eggs and dairy products like milk and cheese, fortified grains, and breakfast cereals, as well as soy and rice milk, and nutritional yeast.

Complete Blood Count (CBC)

Blood Test Results: CBC Explained

A complete blood count (CBC) with differential measures the essential components of the blood including white blood cells, red blood cells, and platelets. Learn more about the various components of a CBC by clicking here.:

<https://www.sonoraquest.com/patients/knowledge-center/knowledge-pages/understanding-the-complete-blood-count-cbc/>

Complete Blood Count (CBC)

A CBC is a commonly ordered blood test that may be helpful to diagnose a variety of health conditions, such as an infection, anemia, or bleeding disorder. It is also a general indicator of overall health. A CBC test screens for many blood components, including red and white blood cells, platelets, and hemoglobin. Learn more about

the various components of a CBC by clicking here.:
<https://www.sonorquest.com/patients/knowledge-center/knowledge-pages/understanding-the-complete-blood-count-cbc/>

Your Results

White Blood Cell Count

Desired Range: 3.8-10.8 Thousand/uL

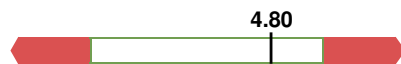


What is White Blood Cell Count?

White blood cells (WBCs) are the body's protectors. Each of the five varieties of WBCs plays its own specific role in defending your body against illness or injury.

Red Blood Cell Count

Desired Range: 3.80-5.10 Million/uL

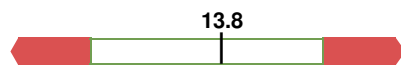


What is Red Blood Cell Count?

Red Blood Cell (RBC) Count - This is a count of the actual number of RBCs per unit of blood. The RBC count is used to aid in the diagnosis of anemia or other conditions that affect red blood cells.

Hemoglobin

Desired Range: 11.7-15.5 g/dL



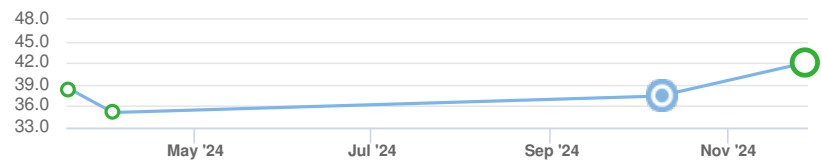
What is Hemoglobin?

Hemoglobin is an iron-containing protein found in red blood cells (RBCs), enabling the cells to carry oxygen and carbon dioxide in the blood. Measuring hemoglobin gives a picture of the ability of the blood to carry oxygen to every cell of your body. A low hemoglobin level may indicate anemia. Hemoglobin increases with altitude adaptation. In general, women have lower hemoglobin values than men.

Hematocrit

Desired Range: 35.0-45.0 %

From 03/17/2024 To 11/28/2024



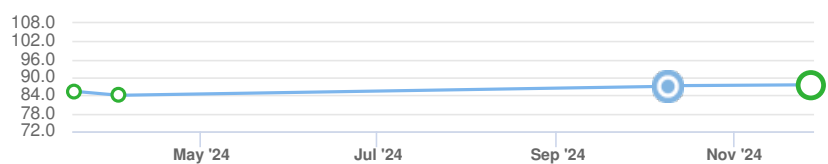
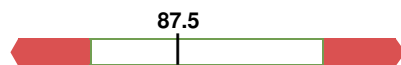
What is Hematocrit?

This screening measures how much of your blood is made of red blood cells (RBCs). A hematocrit measurement is useful in identifying anemia, the presence of liver disease, bleeding disorders and red cell production within the circulatory system. Hematocrit increases with altitude adaptation or dehydration. Women generally have lower hematocrit values than men.

Mean RBC Volume

Desired Range: 80.0-100.0 fL

From 03/17/2024 To 11/28/2024



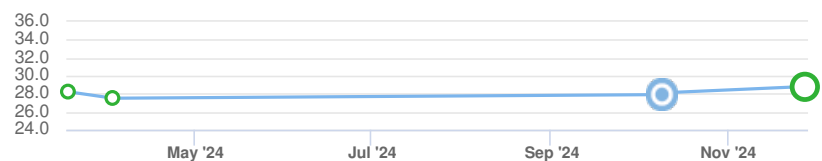
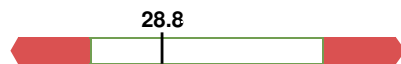
What is Mean RBC Volume?

Mean corpuscular volume (MCV) is a measurement of the average size of your red blood cells (RBCs). A high MCV indicates large, or macrocytic, RBCs. A high MCV is associated with specific varieties of anemia and can indicate a deficiency in vitamin B12 or folic acid. A low MCV indicates small, or microcytic, red blood cells. A low MCV is associated with anemia, and can indicate an iron deficiency, chronic illness or the hereditary disease, thalassemia.

Mean RBC Iron

Desired Range: 27.0-33.0 pg

From 03/17/2024 To 11/28/2024



What is Mean RBC Iron?

Mean corpuscular hemoglobin (MCH) is a calculation of the amount of oxygen-carrying hemoglobin inside your red blood cells (RBCs). A high MCH indicates an increased level of hemoglobin, the oxygen-carrying protein, in your RBCs. Increases can be associated with anemia. Often, increased MCH occurs in conjunction with a high mean corpuscular volume—or large red blood cells—which suggest macrocytic anemia. A low MCH indicates that you have a decreased level of oxygen-carrying protein in your red blood cells. Decreases can be associated with certain types of anemia. Often, decreases occur in conjunction with low mean corpuscular volume—or small RBCs—which suggest microcytic anemia. Hypochromic anemia, resulting from hemoglobin deficiency, can be a possibility as well.

Mean RBC Iron Concentration

Desired Range: 32.0-36.0 g/dL

From 03/17/2024 To 11/28/2024



For adults, a slight decrease in the calculated MCHC value (in the range of 30 to 32 g/dL) is most likely not clinically significant; however, it should be interpreted with caution in correlation with other red cell parameters and the patient's clinical condition.

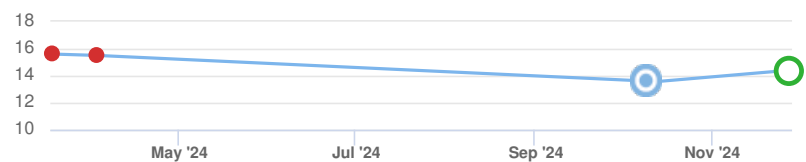
What is Mean RBC Iron Concentration?

Mean corpuscular hemoglobin concentration (MCHC) is the calculation of the percentage of oxygen-carrying hemoglobin in the individual red blood cells (RBCs). A high MCHC indicates an increased level of hemoglobin in your RBCs. Increased MCHC is usually a technical rather than a medical issue. A high level may be caused by distortions in the shape of your RBCs caused by problems in collection, transport or storage of the blood sample. A low MCHC indicates a decreased level of hemoglobin in your RBCs. Decreases can be associated with certain types of anemia, such as iron-deficiency or the hereditary disease, thalassemia.

RBC Distribution Width

Desired Range: 11.0-15.0 %

From 03/17/2024 To 11/28/2024



What is RBC Distribution Width?

Red cell distribution width (RDW) is a calculation of the variation in the size of your red blood cells (RBCs).

Platelets

Desired Range: 140-400 Thousand/uL

From 03/17/2024 To 11/28/2024

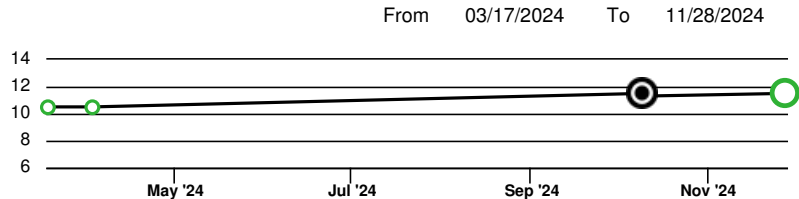


What are Platelets?

Platelets are the smallest type of cell found in the blood. Platelets help stop bleeding after an injury by gathering around the injury site, plugging the hole in the bleeding vessel and helping the blood to clot more quickly. Platelet counts may be done if you are prone to bruising or if you are about to have surgery. The platelet count may change with bleeding disorders, heart disease, diabetes and inflammatory disorders.

MPV

Desired Range: 7.5-12.5 fL

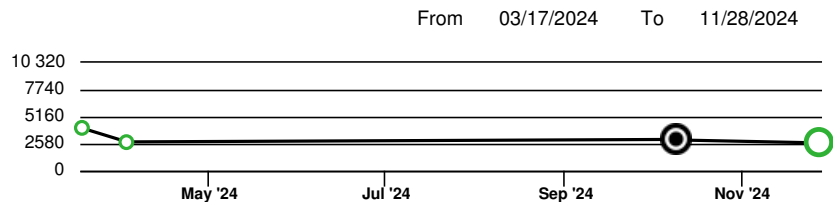


What is MPV?

MPV stands for Mean Platelet Volume. Platelet volume varies with the age of the platelets. Newer platelets are larger than older ones.

Absolute Neutrophils

Desired Range: 1500-7800 cells/uL

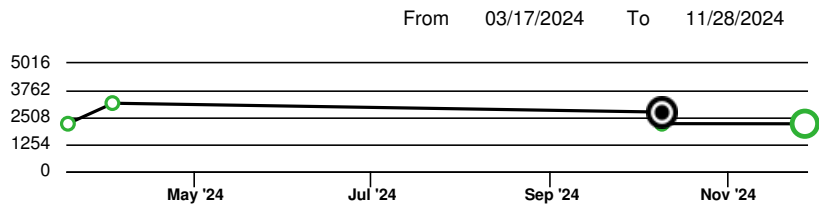
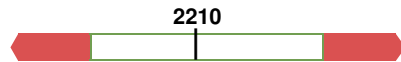


What are Neutrophils?

Neutrophilic granulocytes ("neutrophils") are the most abundant white blood cell. Neutrophils are an essential component of the immune system. They respond to bacterial infections and other types of inflammation. In an infection, neutrophils seep out of the blood vessels in response to factors released as sites of infection. The predominant cells in pus that we observe in a wound are neutrophils.

Absolute Lymphocytes

Desired Range: 850-3900 cells/uL

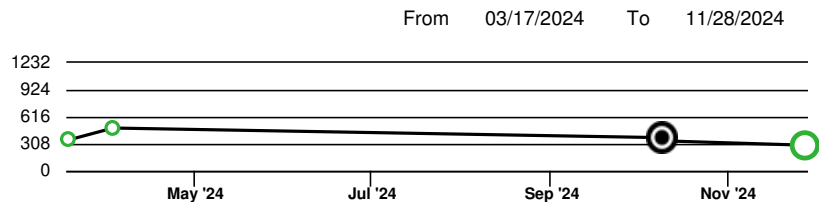


What are Lymphocytes?

Lymphocytes are the second most common white blood cell (WBC). Lymphocytes are divided into larger cells that are also known as natural killer lymphocytes and smaller cells known as B and T lymphocytes. Natural killer cells are important in our immune system to defend against tumors and viral infections. They respond to alterations in the surface of tumor cells and infected cells. B and T lymphocytes adapt to infected cells by either a cellular response mediated by T lymphocytes or antibodies mediated by B lymphocytes.

Absolute Monocytes

Desired Range: 200-950 cells/uL

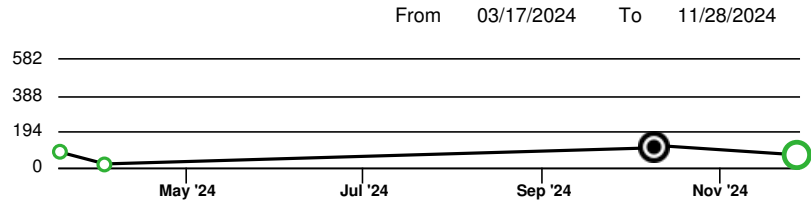


What are Monocytes?

Monocytes, like the other white blood cells (WBCs), originate in the bone marrow, the complex spaces within many of our larger bones. Monocytes are responsible for eating foreign intruders (phagocytosis) and killing infected cells. Monocytes are important in triggering atherosclerosis that affects our arteries and can lead to heart disease and stroke.

Absolute Eosinophils

Desired Range: 15-500 cells/uL

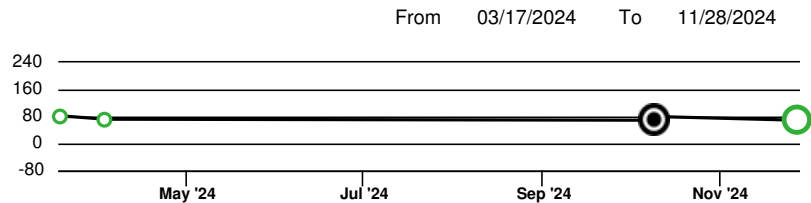
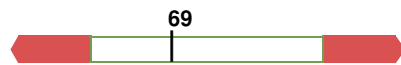


What are Eosinophils?

Eosinophilic granulocytes ("eosinophils") are part of our immune system's response to infection including from parasites. Eosinophils, along with basophils and mast cells, are important in allergic responses and in asthma.

Absolute Basophils

Desired Range: 0-200 cells/uL



What are Basophils?

Basophilic granulocytes ("basophils") - are the least common of the five white blood cell types. When activated, basophils secrete or release many compounds including histamine and interleukin-4. Both compounds are important in the allergic response.

BASIC METABOLIC PANEL

Basic Metabolic Panel

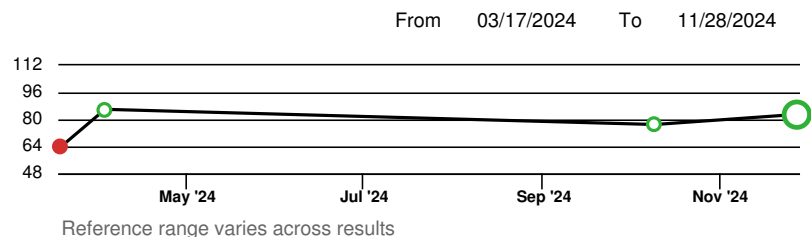
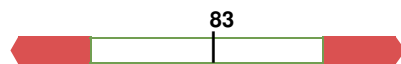
The basic metabolic panel (BMP) is used to check the status of a person's kidneys and their electrolyte and acid/base balance, as well as their blood glucose level. Learn more at the American Association for Clinical Chemistry's (AACC) Lab Tests Online website.

Go to the Lab Tests Online website : <https://labtestsonline.org/understanding/analytes/bmp/tab/test/>

Your Results

Glucose (Blood Sugar)

Desired Range: 65-99 mg/dL



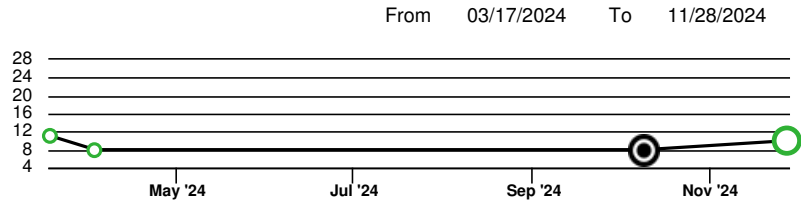
Fasting reference interval

What is Glucose (Blood Sugar)?

Glucose ("blood sugar") is the chief source of energy for all cells in the body. Glucose levels are regulated by hormones produced by your pancreas, including insulin. A glucose level outside the optimal range could be a sign that the body is not correctly producing or using insulin. These conditions are hypoglycemia (low blood sugar), prediabetes (elevated blood sugar), and diabetes (high blood sugar). For the most accurate result you should fast (not eat or drink anything but water) for at least 8 hours before your screening. If you were not fasting at the time of your screening, you should interpret your result against an optimal range of less than 140 mg/dL.

Urea Nitrogen (BUN)

Desired Range: 7-25 mg/dL

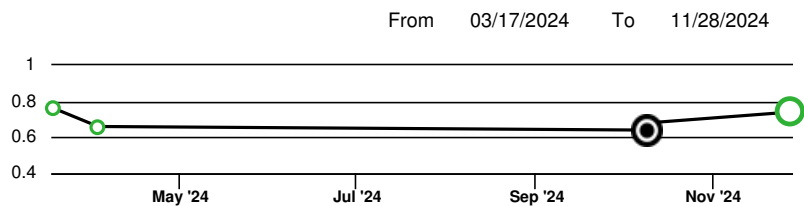
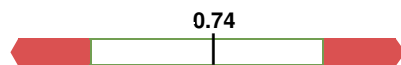


What is Blood Urea Nitrogen (BUN)?

Urea, measured as blood urea nitrogen (BUN) is a waste product derived from the natural breakdown of protein in the liver. Urea is excreted in the urine after blood is filtered through the kidneys. The urea nitrogen level reflects both the metabolism of protein and the effectiveness of the kidneys in filtering blood.

Creatinine

Desired Range: 0.50-0.96 mg/dL

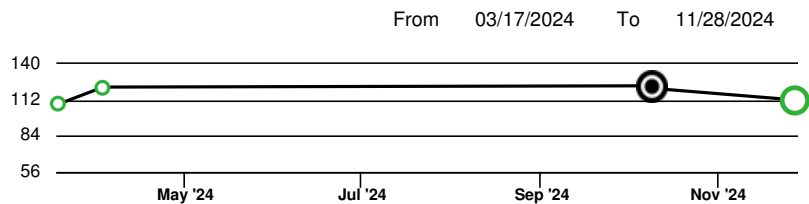


What is Creatinine?

Creatinine is derived from muscles and released into the blood. It is removed from the body by the kidneys. When the creatinine level is elevated, a decrease in kidney function is suggested. For patients 50 years of age and older, and whom identify as African-American the upper reference range for creatinine is approximately 10-15% higher.

EGFR

Desired Range: > OR = 60 mL/min/1.73m²



BUN/Creatinine Ratio

Desired Range: 6-22 (calc)

Result: SEE NOTE:

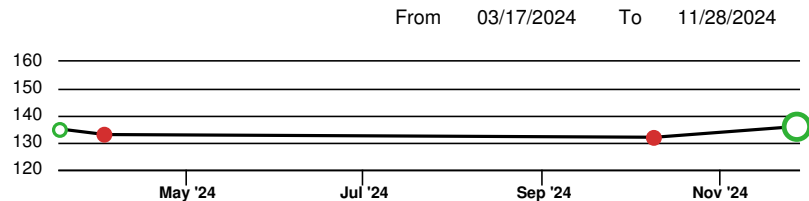
Not Reported: BUN and Creatinine are within reference range.

What is BUN/Creatinine Ratio?

The BUN/creatinine ratio is a calculated value derived by dividing the urea nitrogen result by the creatinine result. This ratio can be helpful in determining whether elevated urea nitrogen is due to impaired kidney function or to other factors such as dehydration, urinary blockage or excessive blood loss.

Sodium

Desired Range: 135-146 mmol/L

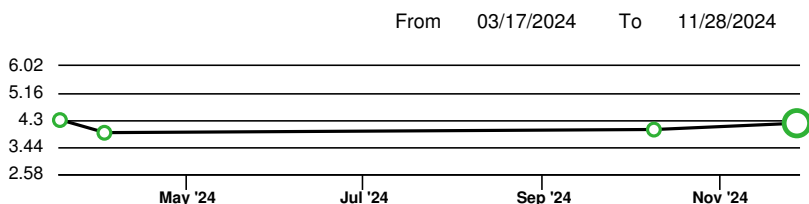
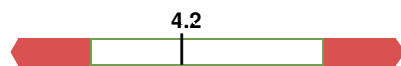


What is Sodium?

Sodium is one of the body's principal minerals, regulated by the kidneys. It plays an important role in water balance in your body. A high level can be caused by dehydration, excessive salt intake in your diet or certain diseases. A low level of sodium may be caused by diarrhea, vomiting, or excessive sweating. Numerous drugs, including diuretics, certain blood pressure medications and steroids, may alter the sodium level.

Potassium

Desired Range: 3.5-5.3 mmol/L



What is Potassium?

What is Potassium?

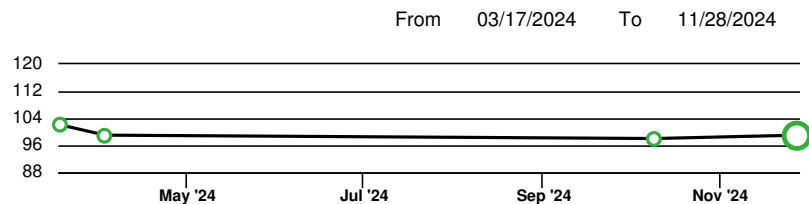
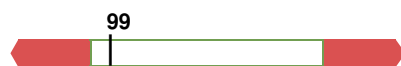
The Potassium Test measures the amount of potassium in your blood. Potassium is an electrolyte that is also one of the body's most important minerals. It helps to regulate water balance, blood pressure, and heart rate. Potassium is also necessary for muscle contraction and nerve function. The relationship between potassium and sodium is essential for proper body function, as their levels need to be balanced for the body to carry out normal functions.

If you don't eat enough potassium, or if your body doesn't store or absorb enough of it, you may have low levels. Too little potassium may cause muscle cramping, muscle weakness, fatigue, irregular heartbeat, constipation, numbness or tingling, or excessive thirst. Too much potassium may cause weak or irregular pulse, irregular heartbeat, nausea, vomiting, chest pain, difficulty breathing, weakness, or fainting. Low or high levels in the blood are of critical significance and should be evaluated by your healthcare provider. This is especially important if you are taking a diuretic or heart medication. A high level may indicate kidney or liver disease, too much medication or bodily injury, such as a burn. A low level of potassium can develop rapidly, most frequently produced as a side effect of drugs that cause increased urination.

Our bodies do not produce potassium, so we must get it from food or supplements. Potassium is naturally found in beets/beet greens, sweet potatoes, tomatoes, white beans, tuna, salmon, cod, broccoli, spinach, avocados, chicken, and some nuts.

Chloride

Desired Range: 98-110 mmol/L

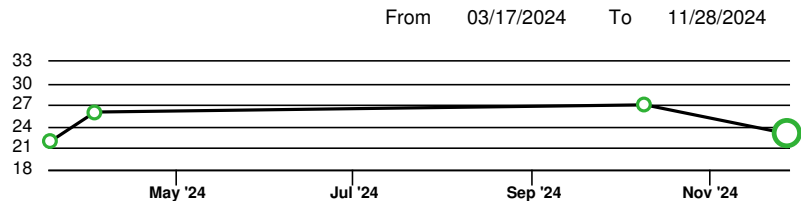


What is Chloride?

Chloride is one of the body's minerals involved with water balance. Most body chloride comes from salt in the diet. A high chloride level may mean severe dehydration, certain kidney disorders or hyperventilation. A low chloride level may result from excessive vomiting, diarrhea, severe burns, excessive sweating or kidney failure. Borderline low or high levels of chloride have very little significance.

Carbon Dioxide

Desired Range: 20-32 mmol/L

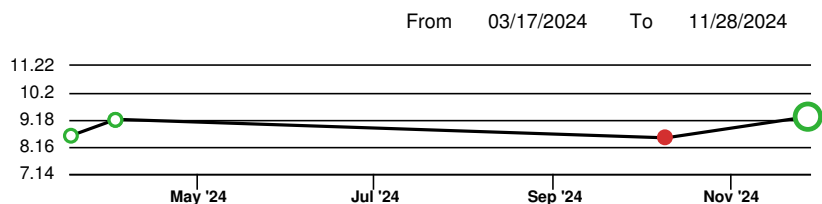
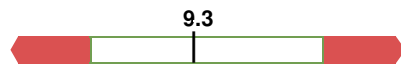


What is Carbon Dioxide?

In the body, most of the carbon dioxide (CO₂) is in the form of a substance called bicarbonate (HCO₃). Therefore, the CO₂ blood test is really a measure of your blood bicarbonate level. Changes in your CO₂ level may suggest that you are losing or retaining fluid, which causes an imbalance in your body's electrolytes. CO₂ levels in the blood are influenced by kidney and lung function. The kidneys are mainly responsible for maintaining the normal bicarbonate levels. The CO₂ level is interpreted with other results to aid in medical diagnoses.

Calcium

Desired Range: 8.6-10.2 mg/dL



What is Calcium?

The Calcium Test measures the total calcium levels in your blood. Calcium is a major mineral the body uses and stores in large amounts. Many people know the importance of calcium for strong bones, but it also helps keep muscles, nerves, and blood vessels working properly. Calcium is necessary for blood clotting, muscular movement, heart function, and bone strength as we age. Ninety-nine percent of the calcium in the body is contained in the bones and only one percent is in the blood. If we don't consume enough of it, our bodies will take it from where we need it the most: our bones. This leads to bone loss and, eventually, osteoporosis (a condition that causes bones to gradually thin and weaken).

For your body to absorb enough calcium, you must also have sufficient levels of vitamin D. Vitamin D promotes calcium absorption and helps to enhance calcium reserves over time. If you don't get eat enough calcium, your body doesn't store or absorb enough of it, your levels may be low. Abnormal calcium levels in the blood may be a sign of bone disease, kidney disorders, thyroid disease, or other conditions.

Our bodies do not produce calcium, so we must get it from food or supplements. Calcium is naturally found in dairy products (milk, cheese, and yogurt), plant milk, fortified orange juice, winter squash, black soybeans, soybeans (edamame), tofu, leafy greens (broccoli, spinach, and kale), fruits, beans, sunflower seeds, chia seeds, and almonds.

HEPATIC FUNCTION PANEL

Hepatic (Liver) Function Tests

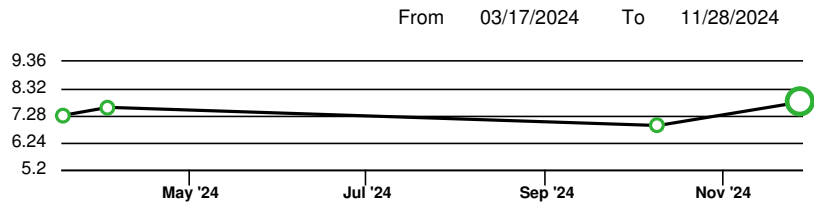
Hepatic or liver function tests are used by your doctor to check your liver's health and detect liver damage. Learn more about these tests from this handout created by the American Liver Foundation (ALF).

Download the handout from the ALF website : <https://liverfoundation.org/wp-content/uploads/2019/11/Liver-Function-Test-Handout-2016.pdf>

Your Results

Protein, Total

Desired Range: 6.1-8.1 g/dL

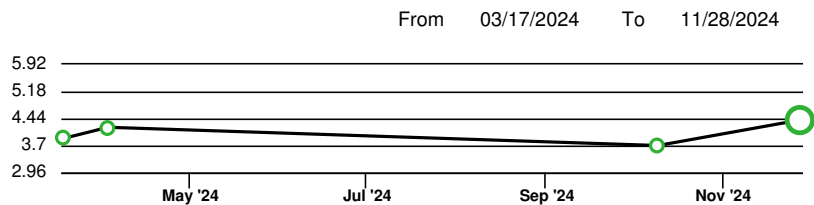
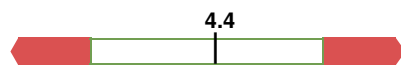


What is Total Protein?

Total protein has two main components—albumin and globulin. The body's protein is derived from ingested food and therefore is influenced by the quality of diet, as well as by liver and kidney function.

Albumin

Desired Range: 3.6-5.1 g/dL



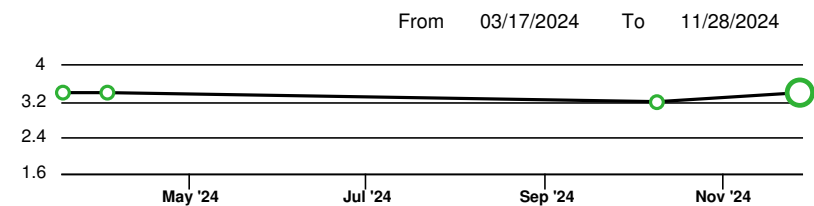
What is Albumin?

Albumin is the largest portion of total blood protein. It transports various metals, drugs, and metabolites throughout the body, including hormones. Like SHBG, it helps in determining the amount of free and bioavailable testosterone in the body.

Decreased blood albumin may indicate many disorders including poor nutrition and advanced liver disease. Modest decreases in albumin may be seen in people with low thyroid gland function and protein-losing conditions.

Globulin

Desired Range: 1.9-3.7 g/dL (calc)

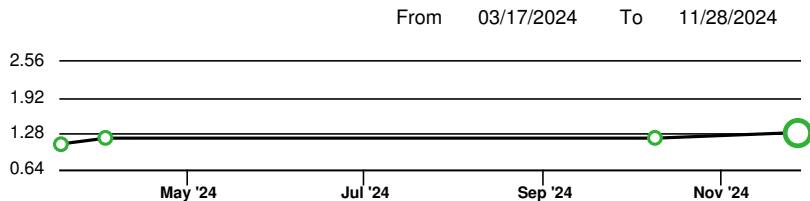


What is Globulin?

Globulin is not measured directly. It is calculated as the difference between the total protein and the albumin levels. The globulins are a group of about 60 different proteins that are part of the immune system, which helps to fight and prevent infections. The globulins also play an important role in blood clotting and serve as carrier proteins for hormones.

Albumin/Globulin Ratio

Desired Range: 1.0-2.5 (calc)

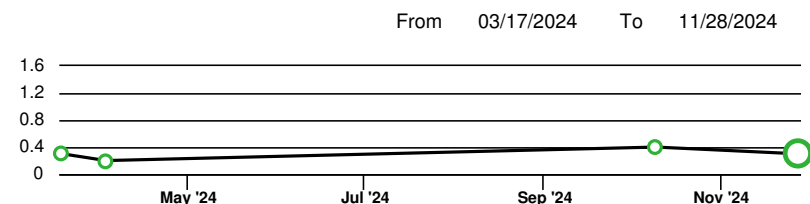


What is Albumin/Globulin Ratio?

Albumin/Globulin Ratio - An alternative way to tell if the albumin or globulin levels in the blood are abnormal is to compare the level of albumin to the level of globulin. If both the albumin and globulin results fall within the specified reference ranges, then a high or low A/G ratio result is not generally considered significant. A high globulin level and low albumin/globulin ratio may suggest high production of globulin that may be due to chronic infections, autoimmune disease, multiple myeloma, and other medical conditions.

Bilirubin, Total

Desired Range: 0.2-1.2 mg/dL

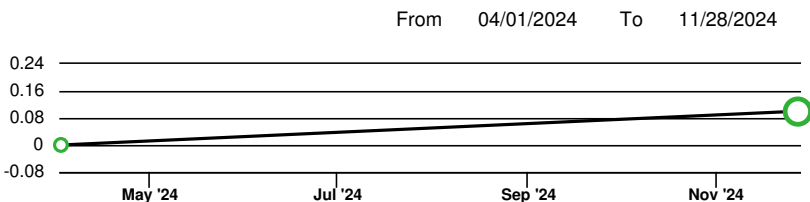


What is Total Bilirubin?

Bilirubin is the main pigment in bile and a major product of normal red cell breakdown. It is helpful in evaluating liver function, various anemias and in evaluating jaundice, yellowing of the skin.

BILIRUBIN, DIRECT

Desired Range: < OR = 0.2 mg/dL

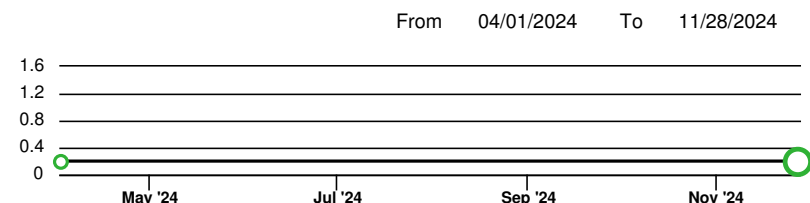


What is BILIRUBIN, DIRECT?

Direct bilirubin is a specific form of bilirubin that is formed in the liver and excreted in the bile. Normally very little of this form of bilirubin is found in the blood. However, in liver disease, this form of bilirubin leaks into the blood so a high level of direct bilirubin may indicate a problem with the liver cells.

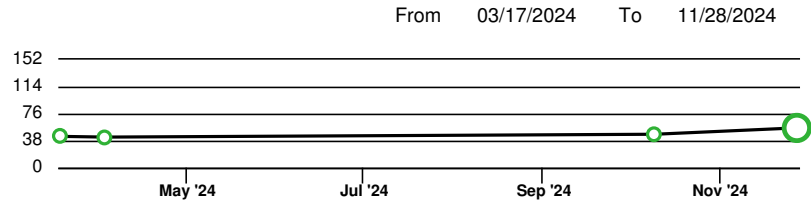
BILIRUBIN, INDIRECT

Desired Range: 0.2-1.2 mg/dL (calc)



Alkaline Phosphatase

Desired Range: 31-125 U/L

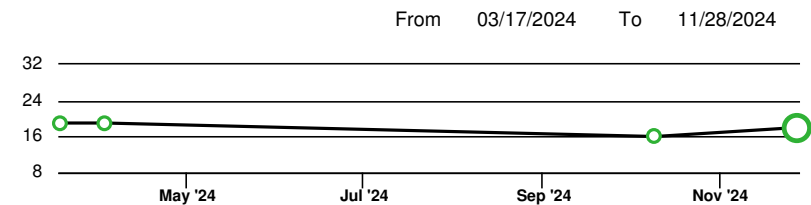
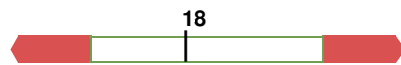


What is Alkaline Phosphatase?

Alkaline phosphatase is an enzyme found primarily in bone and the liver. Elevated levels may indicate the presence of bone or liver disorders. It is commonly increased when the bile duct is blocked which may be caused by gallstones.

Aspartate Aminotransferase

Desired Range: 10-30 U/L

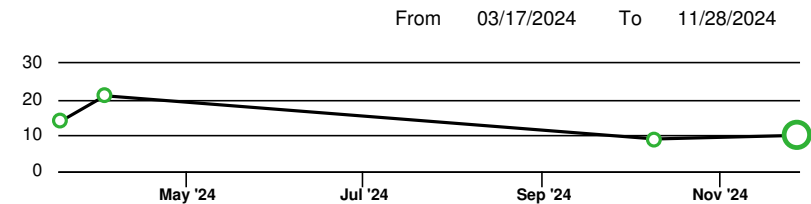


What is Aspartate Aminotransferase?

Aspartate aminotransferase (AST) is an enzyme found in the liver and in cardiac and skeletal muscle. AST may rise in liver, heart, and muscle disorders. It can also rise following strenuous, prolonged exercise.

Alanine Aminotransferase

Desired Range: 6-29 U/L



What is Alanine Aminotransferase?

Alanine aminotransferase (ALT) is an enzyme produced primarily in the liver, skeletal and heart muscle. ALT is present in the liver in a higher concentration than AST and is more specific for differentiating liver injury from muscle damage. ALT rises in the instance of liver disease.

Urine Analysis

Urinalysis Result: What does it mean?

A urinalysis is typically used during a routine physical or when you have symptoms of a UTI, such as abdominal pain, back pain, frequent or painful urination; as part of a pregnancy check-up, a hospital admission, or a pre-surgical work-up.

Your Results

Color

Desired Result: YELLOW

Result: **DARK YELLOW**



What is a Urine Color Test?

This test describes the color of your urine. Many things can affect urine color, including your diet, hydration level, medications and diseases. The color can tell you how much water is in it. Dark yellow has less water and may be a sign of dehydration. Clear, light or pale yellow is a sign of adequate water intake. Some supplements can cause urine to turn bright yellow, green or blue. Some medicines, blackberries, beets, rhubarb, or blood in the urine can turn urine red or brown.

Appearance

Desired Result: CLEAR

Result: **CLEAR**

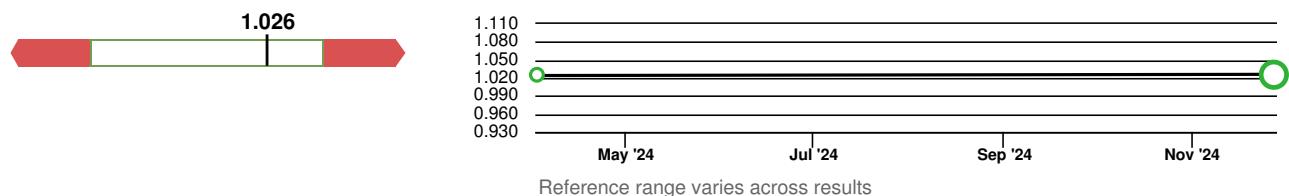


Why is the appearance of urine important?

Urine is normally clear. It can look cloudy when there are bacteria, blood, sperm, crystals, or mucus in the urine.

Specific Gravity

Desired Range: 1.001-1.035

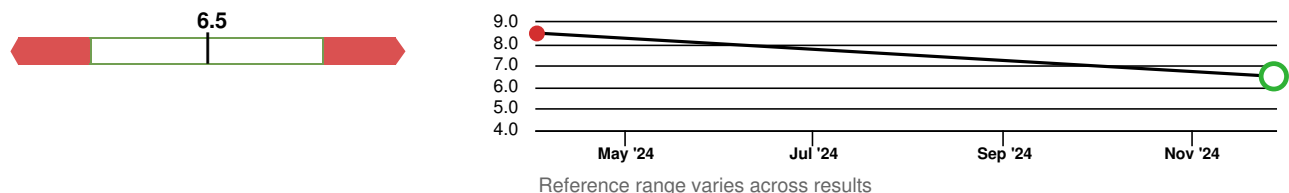


What is Specific Gravity?

Specific gravity looks at the amount of substances in the urine. It can be used to see how well the kidneys are working and as a measure of hydration. When your specific gravity is high, it means there is more solid material in your urine. This often occurs when you are not drinking enough fluids. When specific gravity is low, your kidneys make urine with a high amount of water in it, which is often the result of drinking lots of fluids.

pH

Desired Range: 5.0-8.0



Glucose (Blood Sugar)

Desired Result: NEGATIVE

Result: **NEGATIVE**

NEGATIVE	NEGATIVE
04/02/2024	11/27/2024

What is Glucose (Blood Sugar)?

Glucose ("blood sugar") is the chief source of energy for all cells in the body. Glucose levels are regulated by hormones produced by your pancreas, including insulin. A glucose level outside the optimal range could be a sign that the body is not correctly producing or using insulin. These conditions are hypoglycemia (low blood sugar), prediabetes (elevated blood sugar), and diabetes (high blood sugar). For the most accurate result you should fast (not eat or drink anything but water) for at least 8 hours before your screening. If you were not fasting at the time of your screening, you should interpret your result against an optimal range of less than 140 mg/dL.

Bilirubin

Desired Result: NEGATIVE

Result: **NEGATIVE**

NEGATIVE	NEGATIVE
04/02/2024	11/27/2024

What is Bilirubin?

Bilirubin is a product of red blood cell breakdown. Normally, bilirubin is carried in the blood and passes into your liver, where it's removed and becomes part of bile. Bilirubin in your urine may indicate liver damage or disease.

Ketones

Desired Result: NEGATIVE

Result: **NEGATIVE**

NEGATIVE	NEGATIVE
04/02/2024	11/27/2024

What are Ketones?

When fat is broken down for energy, the body makes substances called ketones (or ketone bodies). These are passed in the urine. Large amounts of ketones in the urine may be as sign of diabetic ketoacidosis, which is a very serious condition. A diet low in carbohydrates, starvation, or severe vomiting may also cause ketones to be in the urine.

Occult Blood

Desired Result: NEGATIVE

Result: **NEGATIVE**

NEGATIVE	NEGATIVE
04/02/2024	11/27/2024

Occult Blood in Urine

Occult blood in urine (also known as microscopic hematuria) is blood that can only be seen with a microscope.

Occult Blood in Urine

Gross hematuria means the urine appears red or the color of tea or cola to the naked eye, whereas occult blood (microscopic hematuria) can only be seen with a microscope.

Causes of both gross and microscopic hematuria include:

- Inflammation of the kidney, urethra, bladder, or prostate (in men)
- Abnormal structures in the urinary tract
- Inherited diseases, like polycystic kidney disease, sickle cell disease, or hemophilia
- Mineral imbalances in the urine
- Glomerulonephritis
- In some cases, no cause of hematuria may be found (idiopathic hematuria)

For more information please visit the National Kidney Foundation site by clicking here: <https://www.kidney.org/atoz/content/hematuria-adults>.

⚠ Protein

Desired Result: NEGATIVE

Result: **TRACE**

NEGATIVE	TRACE
04/02/2024	11/27/2024

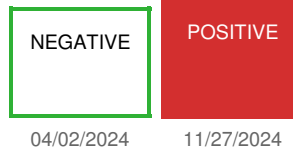
What is Protein?

Protein normally isn't found in the urine. Fever, hard exercise, pregnancy, and some diseases, especially kidney disease, may cause protein to be in the urine.

Nitrite

Desired Result: NEGATIVE

Result: **POSITIVE**



What are Nitrites?

Nitrites are produced when the normally occurring chemical nitrates are converted to nitrites by bacteria that have infected the urinary tract. Nitrites in urine may be a sign of a urinary tract bacterial infection.

Urinary Tract Infections (UTIs)

Urinary Tract Infections (UTIs) are most often caused by bacteria and are common among women. UTIs usually affect the bladder, but also can be present in other parts of the urinary system, including the kidneys. Click here: <https://www.webmd.com/women/your-guide-urinary-tract-infections> to learn more about UTIs.

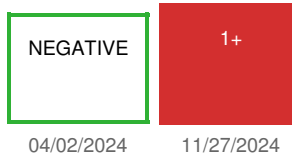
Next Steps

- If you are experiencing any of the following symptoms, contact your healthcare provider immediately:
 - Pain in your upper stomach, lower back, or sides
 - Fever over 100.5 °F
 - Nausea or vomiting
- If you are experiencing any of the following symptoms of a UTI and your test is positive for nitrites, esterase, or the presence of white blood cells, you may call 1.855.205.6146 for a consult with a medical professional who will determine if you may receive a prescription for an antibiotic treatment:
 - Pain in the lower belly area
 - Sudden urges to urinate
 - Burning sensation during urination
 - Urinating more often than usual
- If your test is negative, but you are experiencing symptoms of a UTI (listed above), you can call 1.855.205.6146 to speak to a medical professional who will determine if you may receive a prescription for an antibiotic treatment. A negative urinalysis result does not rule out UTI, but a positive result can help make the diagnosis.

▲ Leukocyte Esterase

Desired Result: NEGATIVE

Result: 1+



What is Leukocyte Esterase?

Leukocyte Esterase shows leukocytes (white blood cells [WBCs]) in the urine. WBCs in the urine may mean a UTI is present.

Urinary Tract Infections (UTIs)

Urinary Tract Infections (UTIs) are most often caused by bacteria and are common among women. UTIs usually affect the bladder, but also can be present in other parts of the urinary system, including the kidneys. Click here: <https://www.webmd.com/women/your-guide-urinary-tract-infections> to learn more about UTIs.

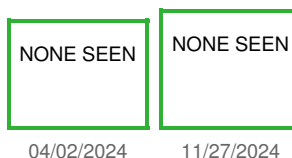
Next Steps

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 - Pain in the lower belly area
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- If your test is negative, but you are experiencing symptoms of a UTI (listed above), you can call 1.855.205.6146 to speak to a medical professional who will determine if you may receive a prescription for an antibiotic treatment. A negative urinalysis result does not rule out UTI, but a positive result can help make the diagnosis.

WBC

Desired Range: < OR = 5 /HPF

Result: NONE SEEN



What is WBC?

Blood cells aren't found in urine normally. Inflammation, disease, or injury to the kidneys, ureters, bladder, or urethra can cause blood in urine. White blood cells (WBCs) may be a sign of infection or kidney disease.

RBC

Desired Range: < OR = 2 /HPF

Result: **NONE SEEN**

NONE SEEN	NONE SEEN
04/02/2024	11/27/2024

What is RBC?

RBCs (female)

The test identifies the presence of red blood cells (RBCs) in the urine. A few RBCs may be present in urine (0-5 RBCs per high power field as viewed with a microscope). Blood in the urine is not a normal finding, but it is not uncommon to have a low level. Excessive RBCs in urine may reflect a medical condition that requires treatment. This test cannot determine the source of the RBCs; therefore, blood from menstrual, vaginal or urinary tract system bleeding cannot be distinguished.

RBCs (male)

The test identifies the presence of red blood cells (RBCs) in the urine. A few RBCs may be present in urine (0-5 RBCs per high power field as viewed with a microscope). Blood in the urine is not a normal finding, but it is not uncommon to have a low level. Excessive RBCs in urine may reflect a medical condition that requires treatment. This test cannot determine the source of the RBCs (i.e., upper or lower urinary tract system).

Squamous Epithelial Cells

Desired Range: < OR = 5 /HPF

Result: **0-5**

0-5	0-5
04/02/2024	11/27/2024

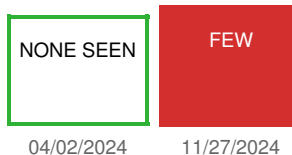
What are Squamous Epithelial Cells?

Squamous epithelial cells are from the skin surface or from the outer urethra. The presence of squamous cells in urine may mean that the urine sample is contaminated. These cells do not mean there is a medical problem.

▲ Bacteria

Desired Range: NONE SEEN /HPF

Result: FEW



What is Bacteria?

This test looks for bacteria in your urine. There are no bacteria in urine normally. If these are present, it can mean you have an infection.

Urinary Tract Infections (UTIs)

Urinary Tract Infections (UTIs) are most often caused by bacteria and are common among women. UTIs usually affect the bladder, but also can be present in other parts of the urinary system, including the kidneys. Click here: <https://www.webmd.com/women/your-guide-urinary-tract-infections> to learn more about UTIs.

Next Steps

- If you are experiencing any of the following symptoms, contact your healthcare provider immediately:
 - Pain in your upper stomach, lower back, or sides
 - Fever over 100.5 °F
 - Nausea or vomiting
- If you are experiencing any of the following symptoms of a UTI and your test is positive for nitrites, esterase, or the presence of white blood cells, you may call 1.855.205.6146 for a consult with a medical professional who will determine if you may receive a prescription for an antibiotic treatment:
 - Pain in the lower belly area
 - Sudden urges to urinate
 - Burning sensation during urination
 - Urinating more often than usual
- If your test is negative, but you are experiencing symptoms of a UTI (listed above), you can call 1.855.205.6146 to speak to a medical professional who will determine if you may receive a prescription for an antibiotic treatment. A negative urinalysis result does not rule out UTI, but a positive result can help make the diagnosis.

Calcium Oxalate Crystals

Desired Range: NONE OR FEW /HPF

Result: FEW

No Historical Data

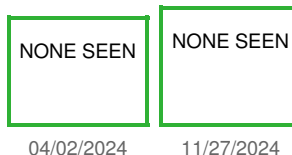
What are Calcium Oxalate Crystals?

Calcium oxalate crystals are the most common crystals in urine. In high amounts, they may be a sign of kidney stones.

Hyaline Cast

Desired Range: NONE SEEN /LPF

Result: NONE SEEN



What is Hyaline Cast?

Hyaline casts are not thought to be indicative of any disease process, but increased numbers may be seen in concentrated urine specimens. Hyaline casts are normal and only indicate that you may have had somewhat concentrated urine at the time it was collected. Highly concentrated urine may be caused by dehydration, exercise or diuretic medicines (water pills).

Note

This urine was analyzed for the presence of WBC, RBC, bacteria, casts, and other formed elements. Only those elements seen were reported.

Vitamin D

The term "vitamin D" refers to different forms of vitamin D. In humans, vitamin D₂ (ergocalciferol) is obtained from foods of plant origin and vitamin D₃ (cholecalciferol) is either obtained from foods of animal origin or produced in our skin after exposure to sunlight. Vitamin D₃ is converted from the inactive form to the active form by action of an enzyme produced in the kidneys. Thus, even mild forms of chronic kidney disease may lead to deficiencies of the active form of vitamin D₃. Vitamin D is essential in maintaining the right levels of calcium and preserving bone strength. Individuals who are deficient in vitamin D are at increased risk of weakening of the bones (osteoporosis) and bone fractures, muscle weakness, high blood pressure, cancer and several forms of autoimmune disorders.

Vitamin D deficiency is becoming increasingly common in India due to diets with inadequate vitamin D and inadequate sunlight exposure. Women who are pregnant have higher needs and are particularly at high risk for vitamin D deficiency and bone loss. Infants who are exclusively breastfed are also subject to vitamin D deficiency. Dietary sources of vitamin D include eggs, fish, and vitamin D fortified milk.

What is vitamin D?

Vitamin D is a fat-soluble vitamin that occurs in 2 forms: vitamin D₃ and vitamin D₂. Vitamin D₃, the more common form, is made in the skin after exposure to sunlight. Vitamin D₂, on the other hand, comes mostly from food and over-the-counter supplements. It may also be used in the pharmacological treatment of vitamin D deficiency.

Vitamin D is rapidly metabolized in the liver to 25-hydroxyvitamin D (25[OH]D). This inactive form is then converted in the kidneys to the active 1,25-dihydroxyvitamin D form.

What does vitamin D do in the body?

Vitamin D helps maintain healthy levels of calcium and phosphorus by aiding in their absorption from the gut. This helps the body form and maintain strong bones. Vitamin D also modulates neuromuscular, immune, and other cellular functions. Vitamin D deficiency has been associated with a wide range of medical conditions including heart disease, hypertension, diabetes, and cancer.

Who should have 25-hydroxyvitamin D testing?

The Endocrine Society recommends screening individuals at risk for deficiency. These include those with ¹:

- Rickets
- Osteomalacia
- Osteoporosis
- Chronic kidney disease
- Hepatic failure
- Malabsorption syndromes
- Hyperparathyroidism
- Medications (anti-seizure medications, glucocorticoids, AIDS medications, antifungals, cholestyramine)

The Society also recommends screening ¹:

- African-American and Hispanic children and adults
- Pregnant and lactating women
- Older adults with history of falls
- Older adults with history of nontraumatic fractures
- Obese children and adults

The Endocrine Society also recommends monitoring patients with granuloma-forming disorders and some lymphomas by testing 25(OH)D and serum calcium. ¹ Some physicians may wish to monitor people receiving vitamin D therapy to evaluate for compliance and expected change in concentration.

How much vitamin D do people need?

To maximize bone health, the Endocrine Society suggests a dietary intake of at least 400 IU/day for **infants <1 year** and at least 600 IU/day for **children 1 year and older**. ¹ Whether these levels are enough to provide all the nonskeletal benefits of vitamin D is not known. At least 1000 IU/day may be needed to raise the blood level consistently above 30 ng/mL (cut point for vitamin D sufficiency). ¹

To maximize bone health and muscle function in **adults 19 to 70 years of age**, the Endocrine Society suggests a dietary intake of at least 600 IU/day. ¹ Whether these levels are enough to provide all the nonskeletal benefits of vitamin D is not known. At least 1500–2000 IU/day may be needed to achieve a blood level of 30 ng/mL. ¹

To maximize bone health and muscle function in **adults over 70 years**, the Endocrine Society suggests a dietary intake of at least 800 IU/day. ¹ Whether these levels are enough to provide all the nonskeletal benefits of vitamin D is not known. At least 1500–2000 IU/day may be needed to achieve a blood level of 30 ng/mL. ¹

Obese children and adults and those on certain medications may need at least 2 to 3 times the suggested dietary intake for their age group. ¹ Relevant medications include anticonvulsants, glucocorticoids, AIDS medications, and antifungals such as ketoconazole.

For people who are vitamin D insufficient or deficient, supplementation or a therapeutic prescription may be needed to correct the deficiency. Refer to the Endocrine Society guidelines ¹ for treatment recommendations.

What are the sources of vitamin D?

Vitamin D can be obtained from exposure to sunlight. However, sun exposure can be affected by season of the year, latitude, time of day, skin pigmentation, use of sunscreens, and age. These variables may necessitate alternative sources for some people.

One alternative source is the diet. Some foods are naturally high in vitamin D; these include oil-rich fish such as salmon, mackerel, and herring. For example, fresh farmed salmon may have approximately 100-250 IU in 3.5 ounces, whereas fresh, wild caught salmon may have approximately 600-1000 IU in a 3.5 ounce serving. Shiitake mushrooms, especially sun-dried, are also high in vitamin D. Other foods are fortified with vitamin D; these include milk and other dairy products, orange juice, and some grain products.

Multivitamin and other supplements are another alternative source.

None of these sources may be adequate for people with liver or kidney disease as they may be unable to produce sufficient amounts of the active form of vitamin D. This is because vitamin D metabolism to the active form requires the liver and kidney. These people may need supplementation with the active form (1,25-dihydroxyvitamin D).

What is the impact of seasons on vitamin D?

25(OH)D concentrations are typically at their lowest at the end of February and at their highest at the end of August. This seasonal effect is more notable in northern latitudes than in southern latitudes where the sun is out for more of the year. Thus, there may be more of a need to supplement, or to supplement with higher doses of vitamin D, in the winter months than in the summer months.

Quest Diagnostics data show that the percentage of patients who are deficient in vitamin D vary seasonally from 21% at the end of summer and 48% at the end of winter.

How common is vitamin D deficiency?

Based on a sample of patients throughout the United States, Quest Diagnostics observed that 33% of patients were deficient in vitamin D, and 60% were either deficient or suboptimal.

What does vitamin D testing measure?

Vitamin D tests generally measure the total concentration of 25(OH)D, which is the main form of vitamin D circulating in blood and the best indicator of vitamin D deficiency or excess. Vitamin D tests using liquid chromatography, tandem mass spectrometry (LC/MS/MS) may also provide the concentration of vitamin D2 and D3 which, when added together, equal the total vitamin D concentration. For detection of vitamin D deficiency, measurement of 1,25-dihydroxyvitamin D is not recommended, as levels may be misleadingly normal in patients with significant vitamin D deficiency.

Why do physicians test for vitamin D?

A physician generally will order a test to determine the level of vitamin D in a patient's body. A physician would typically evaluate the test result in connection with several other factors affecting a patient's health such as medical history, gender, and age.

What are vitamin D2 and vitamin D3?

Vitamin D2 is derived from fungal and plant sources and is commonly found in supplements, such as multivitamins, in the United States. Vitamin D2 may also be used in the pharmacological treatment of vitamin D deficiency. Vitamin D3 is derived from animal sources and is made in the skin following exposure to sunlight. The LC/MS/MS technique is able to directly quantify vitamin D2 and vitamin D3. By comparison, immunoassay-based vitamin D tests can only indirectly measure vitamin D2 and vitamin D3; therefore, only the total vitamin D is reported.

Reference

1. Holick MF, Binkley NC, Bischoff-Ferrari HA, et al. Evaluation, treatment, and prevention of vitamin D deficiency: an Endocrine Society clinical practice guideline. *J Clin Endocrinol Metab.* 2011;96:1911-1930. <https://www.endocrine.org/~media/endosociety/Files/Publications/Clinical%20Practice%20Guidelines/FINAL-Standalone-Vitamin-D-Guideline.pdf>: <https://www.endocrine.org/~media/endosociety/Files/Publications/Clinical%20Practice%20Guidelines/FINAL-Standalone-Vitamin-D-Guideline.pdf>

â This FAQ is provided for informational purposes only and is not intended as medical advice. A physician's test selection and interpretation, diagnosis, and patient management decisions should be based on his/her education, clinical expertise, and assessment of the patient.

Document FAQS.163 Version: 1

Version 1 effective 05/04/2015 to present

Version 0 effective 04/10/2015 to 05/03/2015: <http://education.questdiagnostics.com/faq/FAQ163-retired0>

Vitamin D

Vitamin D is a nutrient that is needed for health and to maintain strong bones by helping the body absorb calcium from food and supplements. Learn more about vitamin D from this fact sheet provided by the NIH's Office of Dietary Supplements.

Download the fact sheet from the NIH website : <https://ods.od.nih.gov/pdf/factsheets/VitaminD-Consumer.pdf>

Vitamin D, 25-Hydroxy

Vitamin D is a nutrient and it acts as a hormone in the body. It can come from some foods, such as salmon and fortified milk, and also supplements, but it is primarily made in the skin after sun exposure. Vitamin D regulates calcium and phosphate levels and is an important regulator of bone mass. It is also involved in actions of the immune system, muscles, and nerves. Vitamin D deficiency can cause osteoporosis (bone thinning and weakness in adults) or rickets, a childhood bone disease.

The liver is the first processor of vitamin D in the body and it metabolizes the fat-soluble vitamin to 25-hydroxyvitamin D, also known as calcifediol. 25-hydroxyvitamin D is used to assess vitamin D levels. More information on vitamin D and vitamin D deficiency can be found at Medline Plus by clicking here:

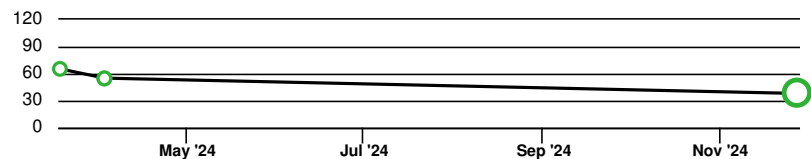
<https://medlineplus.gov/ency/article/003569.htm>.

Your Results

Vitamin D,25

Desired Range: 30-100 ng/mL

From 03/17/2024 To 11/28/2024



Vitamin D Status 25-OH Vitamin D:

Deficiency: <20 ng/mL
Insufficiency: 20 - 29 ng/mL
Optimal: > or = 30 ng/mL

For 25-OH Vitamin D testing on patients on D2-supplementation and patients for whom quantitation of D2 and D3 fractions is required, the QuestAssureD(TM) 25-OH VIT D, (D2,D3), LC/MS/MS is recommended: order code 92888 (patients >2yrs).

What is Vitamin D,25?

What is Vitamin D?

The Vitamin D test measures the amount of 25-OH vitamin D in your blood, an indicator of the level of vitamin D in your body. Vitamin D is an essential fat-soluble vitamin that is necessary for the proper functioning of your heart, blood vessels, insulin, and mood. Vitamin D also helps regulate calcium and phosphorus. These are essential nutrients for strong bones and healthy cells. Vitamin D can also help the immune system fight infections.

If you don't eat enough vitamin D, or if your body doesn't make enough of it, you may have low levels. Not having enough vitamin D can lead to bone pain, muscle weakness, or depression. Vitamin D deficiency can cause osteoporosis (bone thinning and weakness in adults) or rickets, a childhood bone disease.

Your body makes vitamin D after direct exposure to sunlight. It then stores an inactive form of vitamin D in fat cells until it's needed. Your liver and kidneys change the inactive form of vitamin D into the active form (25-hydroxyvitamin D) your body needs. Vitamin D is naturally found in foods such as meat, seafood (trout and salmon), cod liver oil, or fortified dairy and plant-based milk products. It can also be found in supplement form.

COMMENT

See Note 1

Note 1

For additional information, please refer to <http://education.QuestDiagnostics.com/faq/FAQ199> (This link is being provided for informational/ educational purposes only.)

Next Steps

Still have questions about your questhealth.com results?

Contact the PWN care coordination team to schedule a session with a healthcare provider to discuss your results.

Monday - Friday / 9:00am - 5:30pm EST, Phone: (855) 205-6146

NOTE: PWN can only discuss results from tests purchased on questhealth.com

Performing Lab

NL1 Quest Diagnostics LLC-Quest Diagnostics LLC, 200 Forest St, Marlborough, MA 01752-3023 Laboratory Director: Salim E Kabawat M.D.

Ordering Physician

TRUONG,HONG
PWN HEALTH, (888) 362-4321

Key:

MUNGUIA-FLORES,HANNAH (WC620588B)

24 / 25

12/10/24

 Out of Range * **PEND** Pending Result **PRE** Preliminary Result **FINAL** Final Result **RE** Reissued Result

* Refers to a result that is outside of the provided normal range.

Note: Data displayed only for results that meet strict identification matching. Historical result view may vary based on corrected or updated patient demographics. The reference range displayed may vary due to potential changes in laboratory testing methods. Please refer to the published reference range on each lab report.

To feel fully empowered to make healthy decisions based on your test results, it is recommended that you discuss them with a health care provider. Please note, our patient service centers or laboratories do not have your full medical story to provide you with an interpretation or diagnosis.

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